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Simplified Cross-Platform HRM for Puzzle-Solving

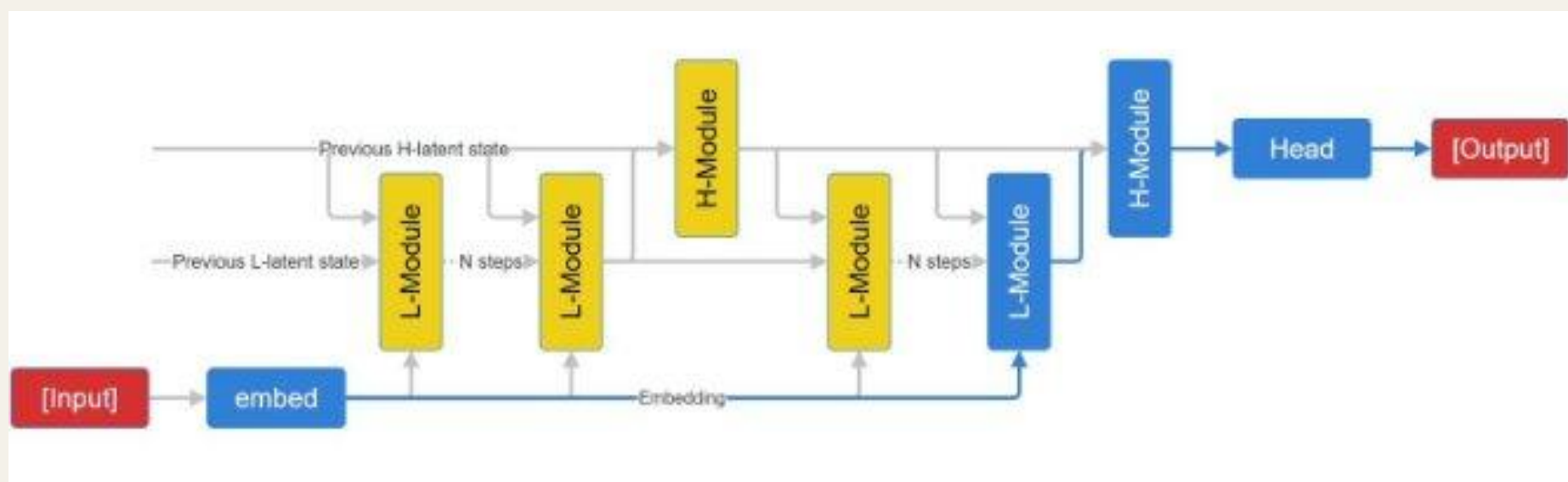
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github.com/fionntmcc/cross-platform-hrm

Introduction

- Reasoning remains a core open challenge in AI. Large Language Models tackle it via Chain-of-Thought (CoT), but errors cascade through the token sequence and fixed-depth Transformers cannot efficiently solve polynomial-time problems. Leading CoT models score 0% on expert Sudoku.
- The Hierarchical Reasoning Model (HRM), introduced by Wang et al. (2025), reasons in continuous latent space through iterative refinement. With only 27M parameters and 1,000 training examples it achieves 40.3% on ARC-AGI-1 and 55% on expert Sudoku.
- Ge et al. (2025) show that a simplified L-module-only variant matches the full HRM while training 2.4x faster, raising the question of whether hierarchical structure is needed at all.

HRM Architecture

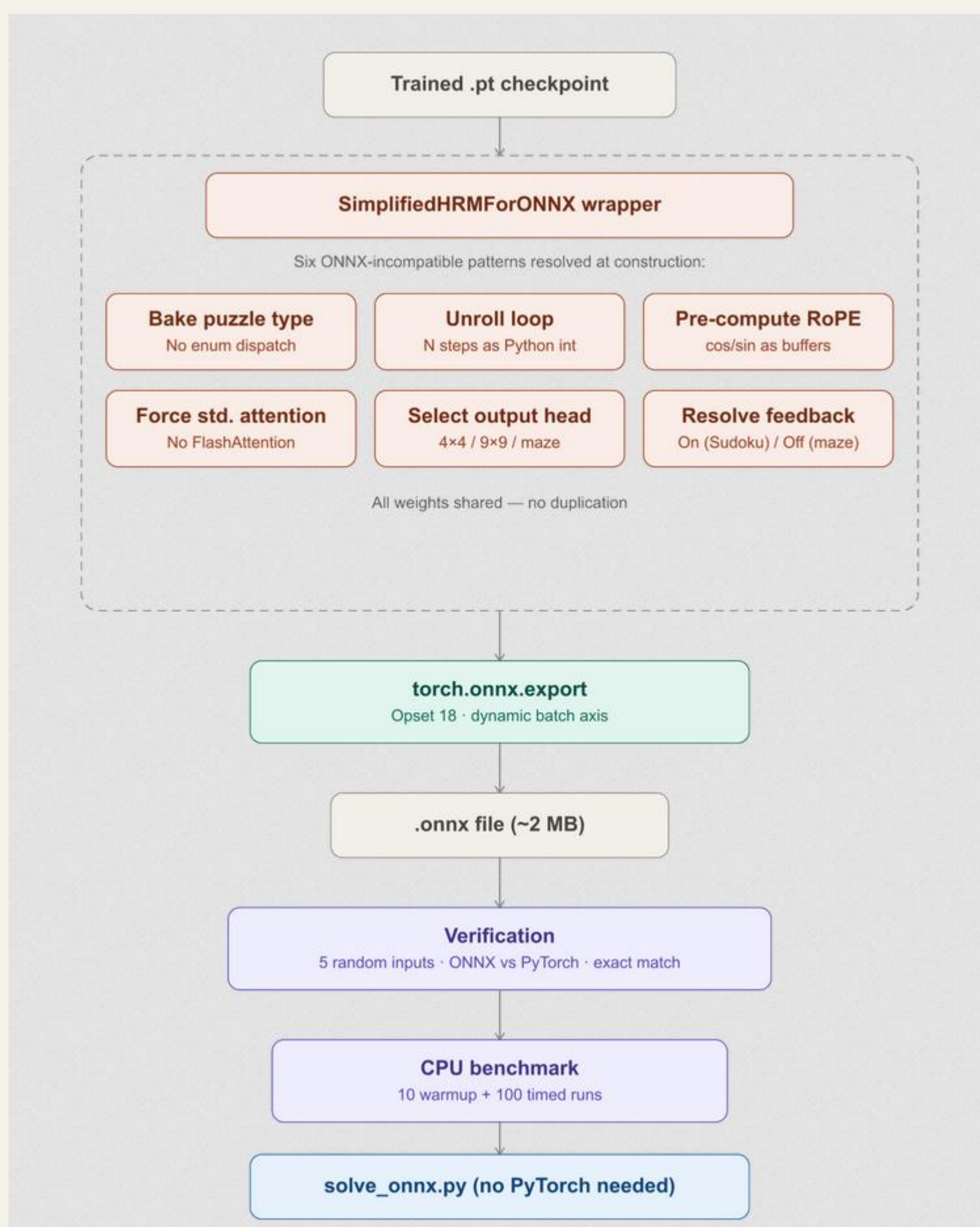


An input embedding feeds an L-module that iterates N times per cycle. After each cycle, the H-module updates once using the converged L-state, resetting the L-module's computational space. One-Step Gradient Approximation gives $O(1)$ training memory versus $O(T)$ for backpropagation through time.

Our Contributions

- Cross-platform HRM implementation in pure PyTorch, no CUDA-specific extensions
- Full HRM and L-module-only (6.9M param) variants implemented from scratch
- Novel Weighted Maze task: first HRM evaluation on cost-sensitive pathfinding
- ONNX export pipeline and zero-dependency Raspberry Pi 5 deployment
- GitHub Actions CI with Black, Ruff, and pytest

ONNX Export Pipeline

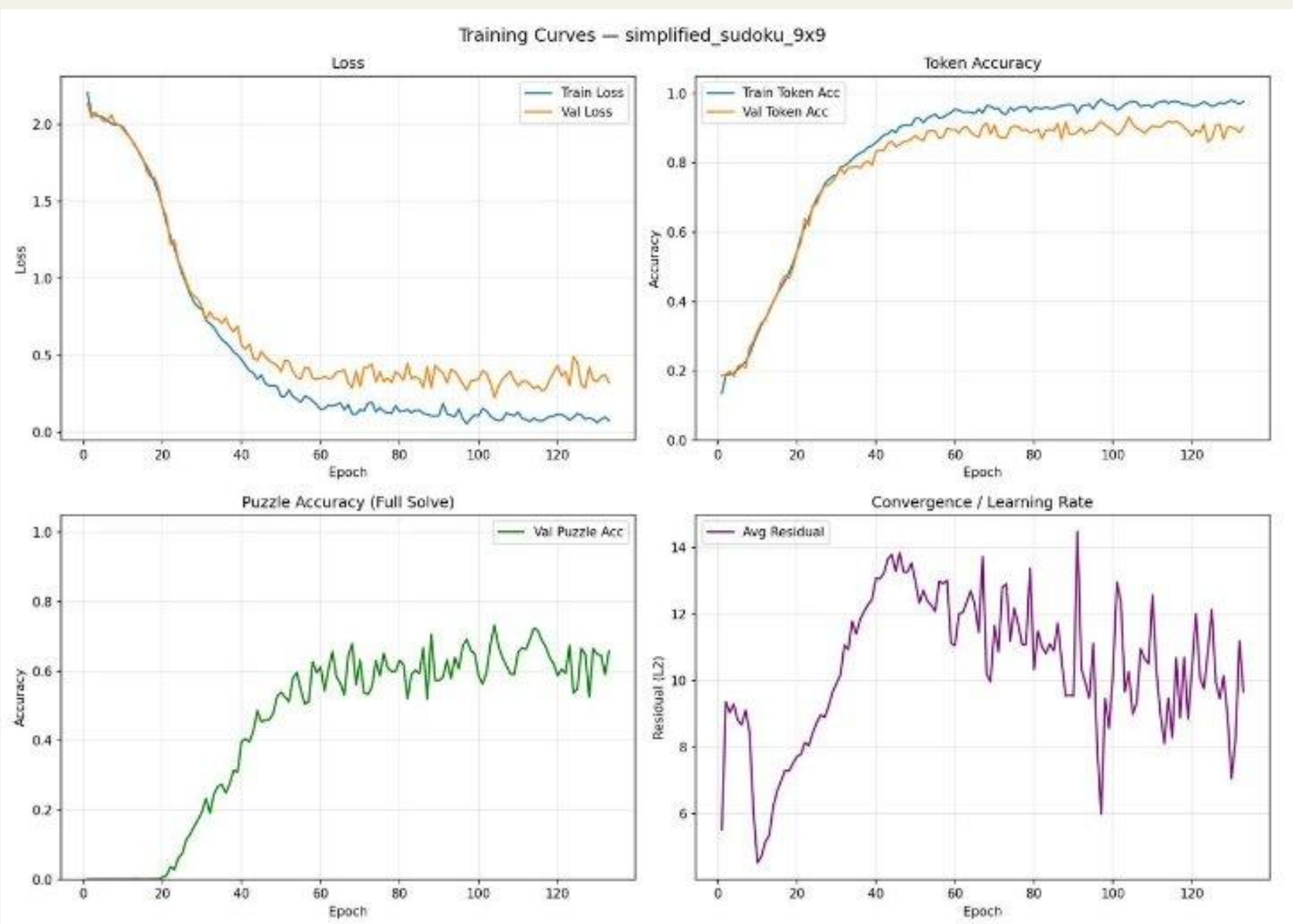


Simplified HRM (L-Module Only)

- Following Ge et al. (2025), we implement a variant that removes the H-module entirely. The L-module consists of 8 weight-shared transformer layers applied iteratively over 16 reasoning steps, with RoPE positional embeddings and SwiGLU-activated MLPs.
- Hidden size 256, 4 attention heads, 6.9M parameters total (vs 27M for the full HRM). Training uses OSGA with deep supervision on a single RTX 5070 Ti GPU (12 GB GDDR7).

Sudoku Results

87.82%



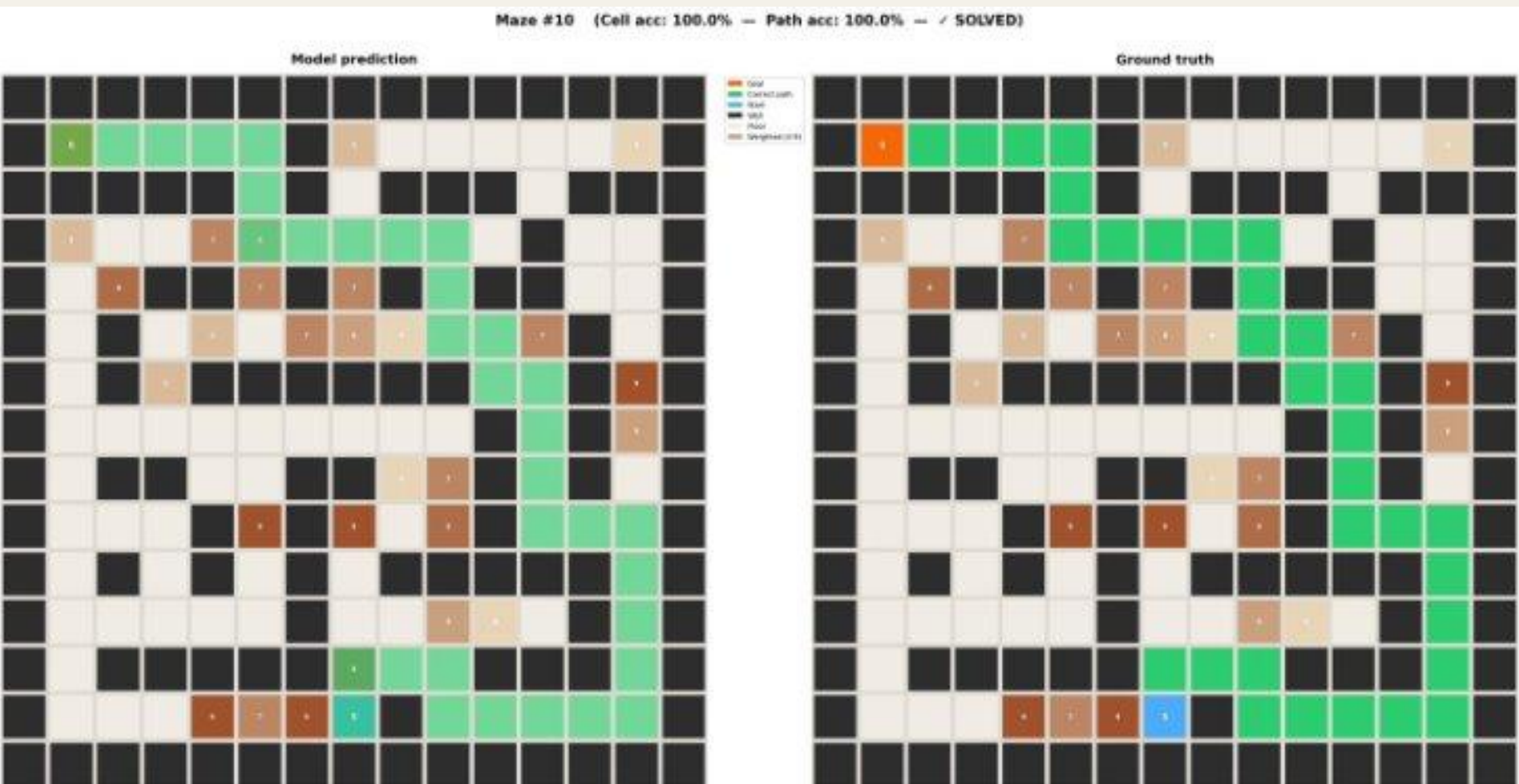
Trained on 10,000 mixed-difficulty 9x9 puzzles spanning easy (<5 backtracks) to expert (50+ backtracks), matching the Sudoku-Extreme distribution. Token accuracy rises rapidly in the first 40 epochs; puzzle accuracy climbs more gradually due to the strict all-or-nothing evaluation criterion.

Novel Task: Weighted Maze

Each traversable cell on a 15x15 grid carries an integer cost from 1 to 9. The model must predict the minimum-cost path from start to goal. Ground truth computed via Dijkstra's algorithm. Not previously evaluated with any HRM variant.

75.20%

Puzzle Accuracy | >98% Cell Accuracy | 15x15 grids | 10 seeds



Trained on 8,000 weighted 15x15 mazes with ground truth paths computed via Dijkstra's algorithm. Cell accuracy exceeds 98%, indicating that even on incorrectly solved mazes the model's predictions are very close to optimal. The gap between cell and puzzle accuracy reflects the strict evaluation: a single misplaced path cell counts the entire maze as failed.

Edge Deployment

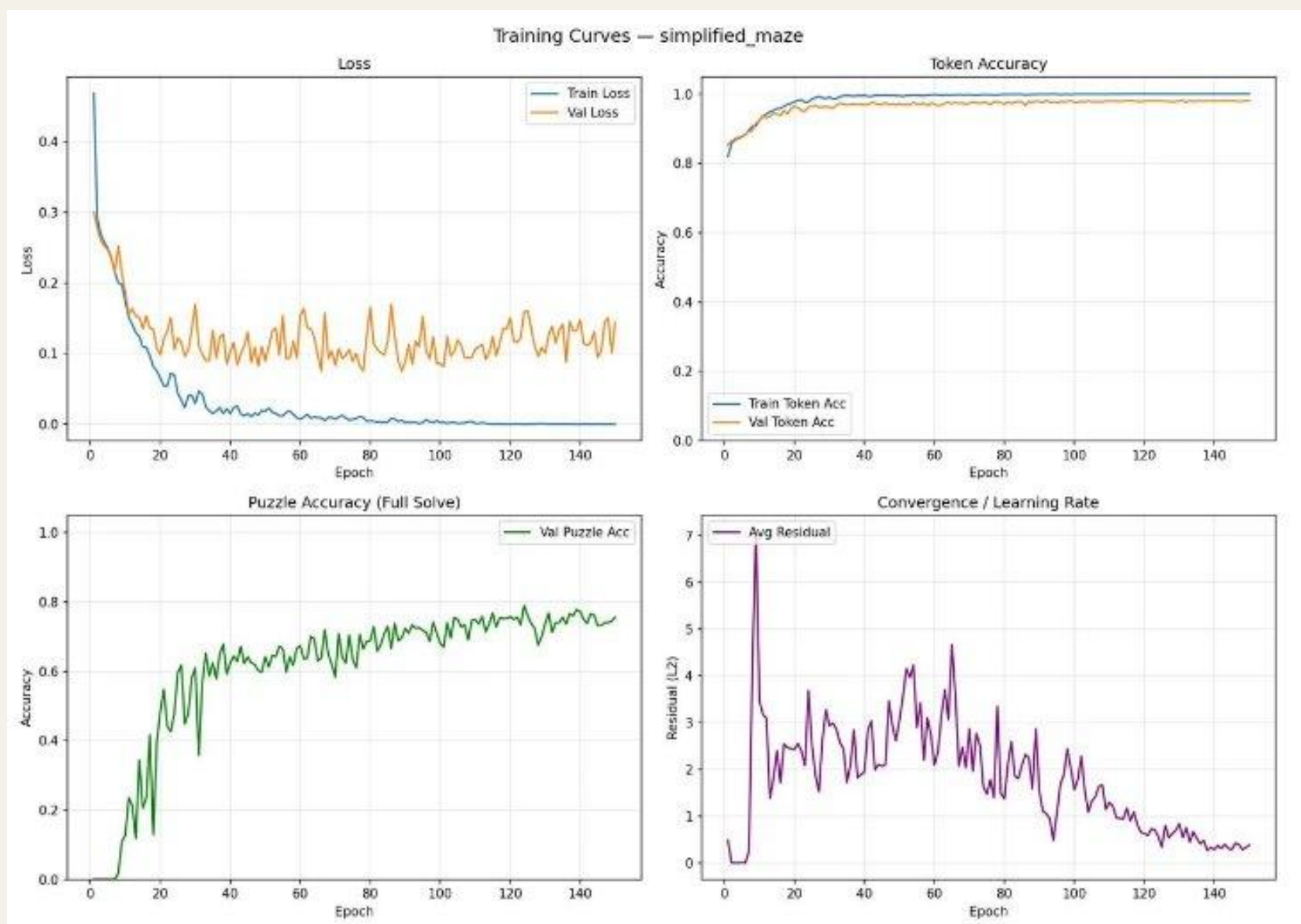
- Model exported to ONNX (opset 18) and deployed on a Raspberry Pi 5 via ONNX Runtime with no PyTorch dependency.
- The SimplifiedHRMForONNX wrapper resolves six ONNX incompatibilities at construction time: baking PuzzleType enum, unrolling the reasoning loop, forcing standard attention off FlashAttention, pre-computing RoPE buffers, inlining InputEmbedding, and selecting the output head.
- Accuracy is identical across GPU, laptop CPU, and Raspberry Pi 5 in FP32, confirming the export preserves iterative reasoning semantics.

Inference Latency

Model	i7-12700H	Pi 5	Slowdown
Sudoku 4x4	13.3 ms	65.2 ms	4.9x
Sudoku 9x9	65.3 ms	343.9 ms	5.3x
Maze 11x11	104.4 ms	521.7 ms	5.0x
Maze 15x15	174.8 ms	1,022.1 ms	5.8x

All benchmarks use 16 reasoning steps, FP32 precision, ONNX Runtime. The Pi 5 is 4.9-5.8x slower than the i7 laptop CPU, with all tasks achieving sub-second latency except Maze 15x15 at 1.02 s.

Weighted Maze Training Curves



Conclusion

The L-module-only variant matches reported HRM reasoning capability at a quarter of the parameter count, supporting the view that computational depth through iterative refinement, not hierarchical structure itself, drives performance. The Weighted Maze task extends HRM evaluation to cost-sensitive pathfinding for the first time. Cross-platform portability from GPU to Raspberry Pi 5 is achievable without architectural compromise, with accuracy preserved across all platforms.

References

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- [2] Ge, R. et al. (2025). Hierarchical Reasoning Models: Perspectives and Misconceptions. arXiv:2510.00355.
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Acknowledgements

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